

You'll learn to build a series of tests and check that each set of inputs results in the output you want. You'll see what a passing test looks like and what a failing test looks like, and you'll learn how a failing test can help you improve your code. You'll learn to test functions and classes, and you'll start to understand how many tests to write for a project.

Installing pytest with pip

While Python includes a lot of functionality in the standard library, Python developers also depend heavily on third-party packages. A *third-party package* is a library that's developed outside the core Python language. Some popular third-party libraries are eventually adopted into the standard library, and end up being included in most Python installations from that point forward. This happens most often with libraries that are unlikely to change much once they've had their initial bugs worked out. These kinds of libraries can evolve at the same pace as the overall language.

Many packages, however, are kept out of the standard library so they can be developed on a timeline independent of the language itself. These packages tend to be updated more frequently than they would be if they were tied to Python's development schedule. This is true of `pytest` and most of the libraries we'll use in the second half of this book. You shouldn't blindly trust every third-party package, but you also shouldn't be put off by the fact that a lot of important functionality is implemented through such packages.

Updating pip

Python includes a tool called `pip` that's used to install third-party packages. Because `pip` helps install packages from external resources, it's updated often to address potential security issues. So, we'll start by updating `pip`.

Open a new terminal window and issue the following command:

```
$ python -m pip install --upgrade pip
```

❶ Requirement already satisfied: pip in /.../python3.11/site-packages (22.0.4)
--snip--

❷ Successfully installed pip-22.1.2

The first part of this command, `python -m pip`, tells Python to run the module `pip`. The second part, `install --upgrade`, tells `pip` to update a package that's already been installed. The last part, `pip`, specifies which third-party package should be updated. The output shows that my current version of `pip`, version 22.0.4 ❶, was replaced by the latest version at the time of this writing, 22.1.2 ❷.

You can use this command to update any third-party package installed on your system:

```
$ python -m pip install --upgrade package_name
```

NOTE

If you're using Linux, `pip` may not be included with your installation of Python. If you get an error when trying to upgrade `pip`, see the instructions in Appendix A.